HelpDesk Application Logic

Key Concepts & User Perspective (No Code)

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1. The Big Picture - What Is This App?

The Problem It Solves

Imagine you work in a company with 100+ employees. Every day:

- Computers break down
- Software doesn't work
- People forget passwords
- Printers jam
- Network issues occur

Without a helpdesk system:

- People send random emails to IT
- Requests get lost or forgotten
- No way to track who's working on what
- No priority system (everything is "urgent!")
- No history of what was done

With a helpdesk system:

- All requests in one place
- Every request gets a unique ticket number
- Clear status tracking (Open → In Progress → Resolved → Closed)
- Priority levels (Low, Medium, High, Critical)
- Accountability (who is assigned to fix it?)
- History and records of all issues

The HelpDesk Solution

```
User has problem → Creates ticket → Support sees it →
Assigns priority → Works on it → Updates status →
Resolves issue → User confirms → Ticket closed
```

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2. User Roles & What They Can Do

Three Types of Users:

```
┌─────────────────────────────────────────────────────────────┐
│                        ADMINISTRATOR                        │
│  • Can do EVERYTHING                                        │
│  • Manages users                                            │
│  • Sees all tickets                                         │
│  • Views system statistics                                  │
│  • Assigns tickets to support staff                         │
│  • Can delete or modify anything                            │
└─────────────────────────────────────────────────────────────┘
                          │
                ┌─────────┴─────────┐
                ▼                   ▼
┌──────────────────────────┐  ┌──────────────────────────┐
│     SUPPORT STAFF        │  │      REGULAR USER        │
│  • Sees assigned         │  │  • Creates tickets       │
│    tickets               │  │  • Views own             │
│  • Updates status        │  │    tickets only          │
│  • Adds comments         │  │  • Adds comments         │
│  • Can reassign          │  │  • Updates profile       │
│  • Views dashboard       │  │  • Can't see             │
```

```
│ • Can't delete │ │ others' tickets │
└──────────────┘ └──────────────────┘
```

Real-World Examples:

Admin (CEO or IT Manager):

- "I need to see all open tickets across the company"
- "I need to create a new support staff account"
- "Let me assign this critical ticket to John"
- "Show me statistics: How many tickets did we resolve this month?"

Support Staff (IT Technician):

- "Show me all tickets assigned to me"
- "Let me update this ticket to 'In Progress'"
- "I'll add a comment: 'Replaced keyboard, testing now'"
- "This is fixed, marking as Resolved"

Regular User (Employee):

- "My computer won't turn on, I need help!"
- "Let me check if my previous ticket was resolved"
- "I'll reply to the support person's question"
- "Yes, it's working now, thanks!"

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3. The Ticket Lifecycle

The Journey of a Ticket:

```
1. OPEN (Created)
   └─> User creates ticket: "My printer won't print"
   └─> Status: OPEN (red/orange)
   └─> Assigned to: Nobody yet

2. ASSIGNED
   └─> Admin sees it and assigns to John (Support)
   └─> Status: Still OPEN
```

```
   └─> Assigned to: John
   └─> John gets notification

3. IN PROGRESS (Being worked on)
   └─> John starts working on it
   └─> Status: IN PROGRESS (yellow/blue)
   └─> John adds comment: "Checking the printer now"

4. RESOLVED (Fixed)
   └─> John fixes the issue
   └─> Status: RESOLVED (green)
   └─> John adds comment: "Replaced toner cartridge, printer works now"
   └─> User gets notification

5. CLOSED (Confirmed fixed)
   └─> User confirms it works
   └─> Status: CLOSED (gray)
   └─> Ticket archived
```

Visual Status Flow:

```
┌─────────┐    ┌──────────────┐    ┌──────────┐    ┌─────────┐
│ OPEN    │───>│ IN PROGRESS  │───>│ RESOLVED │───>│ CLOSED  │
└─────────┘    └──────────────┘    └──────────┘    └─────────┘
                  │                  │
                  │                  │
                  └──────────────────────────────────┘
                  (Can be reopened if problem returns)
```

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4. User Workflows & Journeys

Workflow A: Regular User Creates a Ticket

Scenario: Jane's computer screen is flickering

Steps Jane Takes:

1. **Login**
   - Opens app: `http://172.25.6.233:3001`
   - Enters username: `jane.smith`

- Enters password: `password123`
- Clicks "Login"

2. **Dashboard View**
   - Sees her previous tickets (if any)
   - Sees a button: "Create New Ticket"
   - Clicks it
3. **Create Ticket Form**
   - **Title**: "Computer screen flickering"
   - **Description**: "My monitor keeps flickering. It started this morning. Sometimes goes black for a few seconds."
   - **Priority**: She selects "High" (can't work properly)
   - Clicks "Submit"
4. **Confirmation**
   - Gets ticket number: "HD-12345678"
   - Status shown as: "Open"
   - Message: "Your ticket has been submitted. Our support team will review it soon."
5. **Waiting**
   - Jane can see her ticket in "My Tickets"
   - Status: Open
   - She can add more comments if needed
6. **Gets Update**
   - Support person (John) adds comment: "Hi Jane, I'll come check your monitor at 2 PM"
   - Jane sees notification
   - She can reply: "Thank you! I'll be at my desk."
7. **Resolution**
   - John marks ticket as "Resolved"
   - Jane receives notification
   - She confirms it's working
   - Ticket automatically closes

Workflow B: Support Staff Handles Ticket

Scenario: John (Support) receives a ticket assignment

Steps John Takes:

1. **Login**
   - Logs in as: `john.doe`
   - Role: Support
2. **Dashboard View**
   - Sees metrics:
     - "5 tickets assigned to you"
     - "2 high priority"
     - "3 open tickets"
   - Sees list of assigned tickets sorted by priority
3. **Reviews Tickets**
   - Clicks on high priority ticket
   - Reads: "Computer screen flickering"
   - Sees who created it: Jane Smith
   - Sees when: 9:15 AM today
4. **Takes Action**
   - Changes status: Open → In Progress
   - Adds comment: "Hi Jane, I'll come check your monitor at 2 PM"
   - Clicks "Update"
5. **Works on Issue**
   - Goes to Jane's desk
   - Discovers HDMI cable is loose
   - Fixes it
6. **Updates Ticket**
   - Adds comment: "Fixed - The HDMI cable was loose. Secured it properly. Please test and let me know if the flickering continues."
   - Changes status: In Progress → Resolved
   - Clicks "Update"
7. **Waits for Confirmation**

- Jane confirms it's working
- Ticket auto-closes
- John moves to next ticket

Workflow C: Admin Manages System

Scenario: Admin needs to monitor helpdesk performance

Steps Admin Takes:

1. **Login as Admin**
   - Username: `admin`
   - Full access to everything
2. **Dashboard Overview**
   - Sees company-wide statistics:
     - Total tickets: 145
     - Open: 23
     - In Progress: 12
     - Resolved this week: 45
     - Average resolution time: 4.5 hours
   - Sees graphs and charts
3. **Manages Unassigned Tickets**
   - Filters: "Show unassigned tickets"
   - Sees 8 tickets waiting
   - Assigns based on:
     - Priority (High → Senior staff)
     - Type (Network issue → Network specialist)
     - Workload (Who has fewer tickets?)
4. **Assigns Ticket**
   - Opens ticket "HD-12345679: WiFi not working"
   - Priority: Critical
   - Assigns to: John (Network specialist)
   - John immediately gets notification

5. **User Management**
    - Goes to "Users" section
    - Sees all users:
        - 3 Admins
        - 8 Support staff
        - 150 Regular users
    - Can add new support staff
    - Can deactivate users who left company
6. **Reviews Performance**
    - Which support staff resolved most tickets?
    - What types of issues are most common?
    - Any tickets taking too long?

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5. Key Business Logic Rules

Rule 1: Who Can See What?

```
REGULAR USER:
✅ Can see: Only their own tickets
❌ Cannot see: Other users' tickets
❌ Cannot see: All users list
❌ Cannot see: System statistics

SUPPORT STAFF:
✅ Can see: Tickets assigned to them
✅ Can see: All open tickets (to volunteer)
✅ Can see: Basic statistics (their performance)
❌ Cannot see: Other support staff's tickets (unless reassigned)
❌ Cannot see: User management

ADMIN:
✅ Can see: EVERYTHING
✅ Can do: EVERYTHING
```

Rule 2: Status Transitions

What transitions are allowed?

```
Open → In Progress ✅ (Support starts working)
Open → Resolved ✅ (Quick fix, no progress step needed)
In Progress → Resolved ✅ (Support finishes)
Resolved → Closed ✅ (User confirms OR auto-closes after 24 hours)
Resolved → Open ✅ (Problem came back!)
Closed → Open ✅ (Problem returned, reopen)

NOT ALLOWED:
Open → Closed ❌ (Must be resolved first!)
In Progress → Closed ❌ (Must be resolved first!)
Closed → In Progress ❌ (Must reopen first!)
```

Rule 3: Priority Levels

How to decide priority:

```
CRITICAL (Red):
- System is completely down
- Affects multiple users
- Business-stopping
- Example: "Entire office has no internet"
- Response time: Immediate

HIGH (Orange):
- User cannot work
- Single-user impact but severe
- Example: "Computer won't turn on"
- Response time: Within 2 hours

MEDIUM (Yellow):
- User can work but with difficulty
- Not urgent but important
- Example: "Printer jammed"
- Response time: Same day

LOW (Green):
- Minor inconvenience
- Can wait
- Example: "Mouse wheel squeaky"
- Response time: Within 3 days
```

Rule 4: Ticket Assignment

Who gets assigned?

```
AUTOMATIC RULES:
1. Load balancing: Assign to person with fewest tickets
2. Expertise: Network issues → Network specialist
3. Availability: Don't assign to people on vacation
4. Priority: Critical tickets → Senior staff

MANUAL OVERRIDE:
Admin can assign to anyone, overriding rules
```

Rule 5: Notifications

When are notifications sent?

```
USER RECEIVES NOTIFICATION WHEN:
- Ticket is assigned to support staff
- Support adds a comment
- Status changes to "Resolved"
- Support asks a question

SUPPORT RECEIVES NOTIFICATION WHEN:
- New ticket is assigned to them
- User adds a comment to their ticket
- Admin reassigns ticket to them
- Priority is upgraded to Critical

ADMIN RECEIVES NOTIFICATION WHEN:
- Critical priority ticket is created
- Ticket is unresolved for > 24 hours
- User complains or escalates
```

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6. Data Flow & Relationships

How Data Connects:

```
USER
 │
 ├─ creates ──> TICKET
 │              │
```

```
│           ├─ assigned to ──> SUPPORT STAFF
│           │
│           ├─ has many ──> COMMENTS
│           │           (from user & support)
│           │
│           └─ has ──> STATUS
│                   (Open, In Progress, etc.)
│
└─ has ──> PROFILE
        (name, email, picture)
```

Example with Real Data:

```
TICKET #HD-12345678
├─ Title: "Printer won't print"
├─ Description: "Getting error message when I try to print"
├─ Status: In Progress
├─ Priority: Medium
├─ Created by: Jane Smith (User #4)
├─ Assigned to: John Doe (Support #3)
├─ Created at: 2026-01-13 09:30 AM
├─ Updated at: 2026-01-13 10:15 AM
└─ Comments:
   ├─ [09:30 AM] Jane: "Tried restarting, still doesn't work"
   ├─ [09:45 AM] John: "I'll check it in 10 minutes"
   └─ [10:15 AM] John: "Paper was jammed. Fixed now. Please try."
```

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7. Real-World Scenarios

Scenario 1: The Overwhelmed Support Person

Problem:

- John has 15 open tickets
- 3 are marked Critical
- He's working alone today

What happens:

1. **Admin notices:** Dashboard shows John is overloaded

2. **Admin action:** Reassigns 5 tickets to Sarah (another support)
3. **Automatic notification:** Sarah gets notified of new assignments
4. **Result:** Workload balanced, critical tickets handled faster

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Scenario 2: The Recurring Problem

Problem:

- Same user reports same issue 3 times in 1 week
- Ticket keeps getting reopened

What happens:

1. **Pattern detected:** Admin sees ticket has been reopened twice
2. **Root cause analysis:** Not just a quick fix needed
3. **Escalation:** Admin assigns to senior tech
4. **Permanent fix:** Replace equipment instead of patching
5. **Documentation:** Add comment explaining permanent solution

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Scenario 3: The Critical Emergency

Problem:

- Server room temperature alarm goes off
- Could damage all servers!

What happens:

1. **Ticket created:** Automatically marked Critical
2. **Multiple notifications:** All admins + facility staff notified
3. **Immediate assignment:** Assigned to on-call engineer
4. **Status updates:** Every 10 minutes until resolved
5. **Post-mortem:** After resolution, document what happened and how to prevent

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Scenario 4: The Happy Path

Problem:

- Simple password reset request

Flow:

1. **User:** "I forgot my password"
2. **Ticket:** Created, marked Low priority
3. **Support:** Sees it within 30 minutes
4. **Action:** Resets password, sends new one
5. **Resolution:** Completed in 5 minutes
6. **Closure:** User confirms, ticket closed
7. **Total time:** 35 minutes from creation to closure

Key Concepts Summary

1. Accountability

Every ticket has an owner (assigned person). Nothing falls through cracks.

2. Transparency

Everyone knows the status. No mystery about what's happening.

3. Priority Management

Critical issues get handled first. Not "first come, first served."

4. Documentation

Comments create a history. Next time, support can see what was tried before.

5. Metrics & Improvement

Track everything: response times, resolution times, common issues.
Use data to improve service.

6. Separation of Concerns

- Users focus on: Describing their problem
- Support focuses on: Solving problems

- Admins focus on: System efficiency

7. Communication Trail

All communication stays in the ticket. No lost emails or forgotten conversations.

The Mental Model

Think of the HelpDesk app as a **smart notebook** where:

- Every page = A ticket
- Page header = Ticket number, status, priority
- Page content = Description + conversation thread
- Sticky notes = Status updates
- Assignment stamp = Who's responsible
- Bookmark color = Priority level
- Archive = Closed tickets

But smarter than a notebook:

- Automatic notifications
- Can't lose pages
- Searchable
- Shows statistics
- Role-based access
- Tracks everything

User Experience Goals

For Regular Users:

- "I can easily report my problem"
- "I know someone is working on it"
- "I get updates without asking"
- "I can check status anytime"

For Support Staff:

- "I see my workload at a glance"
- "I know what to work on first"
- "I can track my progress"
- "I can communicate with users easily"

For Admins:

- "I see the big picture"
- "I can balance workload"
- "I can spot patterns"
- "I can measure performance"

Next Steps

Now that you understand:

- **WHAT** the app does (manages support tickets)
- **WHO** uses it (Admin, Support, Users)
- **HOW** it flows (Ticket lifecycle)
- **WHY** decisions are made (Business rules)

You're ready to understand the **CODE** that makes this happen!

The logic you've learned here translates directly into:

- Database design (how data is stored)
- API endpoints (how data is accessed)
- Frontend components (how users interact)
- Business logic (rules enforced in code)

Would you like me to explain the code implementation next? 🚀